THE UNIVERSITY OF DA NANG

VNUK INSTITUTE

FOR

RESEARCH AND EXECUTIVE EDUCATION

Human & Computer Interaction Project

-

Automated Investment Platform

By

Phan Van Quy 21020003

Truong Ha Vu 20020001

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PART 1 - ESSAY

I/ Brief:

Background:

- Automated investment strategies help investors maximize returns with minimal manual effort, especially in decentralized finance (DeFi).
- The project, "Developing an Automated Investment Platform on DEX," focuses on creating a user-friendly dollar-cost averaging (DCA) solution for decentralized exchanges (DEXs).
- Users can create and manage DCA pools ("machines") to automate token conversions over time.
- The DCA approach reduces risks from market volatility and supports long-term, consistent growth.
- This platform simplifies investments, catering to users with limited time or expertise in portfolio management.

A. Project: Auto Investment on DEX

B. Brand: AutoDEX

C. Client: DeFi enthusiasts and digital asset investors

D. Project Descriptions:

- Brand Introduction
- Create Investment Machines

- Manage Investment Strategies
- Real-Time Monitoring
- Secure Asset Management
- User-Friendly Interface
- Automated Investment Execution

E. Budget: N/A

F. Brand Background:

- Founded 2024 by WalWee & HaVux with the vision of revolutionizing DeFi investment automation.
- Motto: "Invest smarter, grow consistently."

G. Target Audience:

- DeFi enthusiasts
- Long-term crypto investors
- Individuals seeking secure, automated investment solutions

H. Coverage: Global audience of DeFi participants, both novice and experienced, looking for transparency and advanced investment tools.

I. Timing: N/A

J. Description:

- **Web:** Solidity, Web3.js
- **Smart Contracts:** Secure, decentralized protocols

K. Objective:

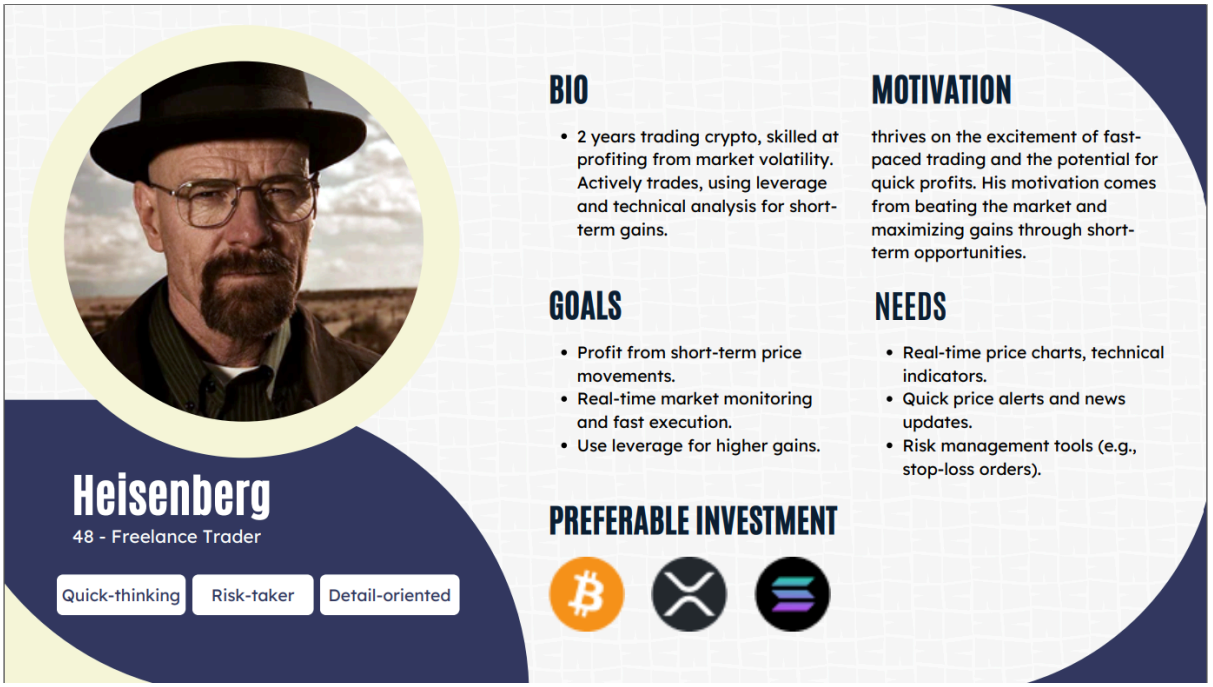
- Build a platform that simplifies and automates investment strategies in the DeFi ecosystem.
- Provide advanced tools like DCA machines for consistent and risk-managed growth.
- Ensure user trust with high security and a streamlined user experience.

II/ Analyze:

A. Scenarios:

- **Step 1:** Walter, a crypto enthusiast, decides to automate his investment strategy using the AutoDEX application.
- **Step 2:** Walter installs the app on his smartphone and opens it. Since he hasn't connected a wallet yet, he is directed to the Connect screen.
- **Step 3:** Walter taps the Connect button, which redirects him to the MetaMask app for authentication. Once the wallet is successfully connected, he is redirected back to AutoDEX.
- **Step 4:** Navigating to the Strategy screen, Walter creates a new Machine by setting parameters like token type, investment amount, and frequency. After confirming, the app prompts him to sign the transaction in his wallet.
- **Step 5:** The Machine is created and stored in the system. Walter can view and manage it on the My Machines screen, where he monitors its performance and adjusts settings if needed.
- **Step 6:** On the Profile screen, Walter checks his overall portfolio, analyzing asset distribution and performance to ensure his investments align with his goals.

B. Personas:



Overall description:

- An experienced freelance crypto trader with two years in cryptocurrency trading.
- Thrives in the fast-paced environment of short-term investments, leveraging his quick-thinking and detail-oriented nature to capitalize on market volatility.
- He employs technical analysis and advanced trading strategies, including the use of leverage, to maximize returns within short timeframes.
- His preference lies in decentralized, high-liquidity tokens like Bitcoin, XRP, and Solana.

Goals:

- Consistently profit from short-term market fluctuations.
- Monitor real-time market trends and execute trades swiftly to stay ahead of competitors.
- Minimize risk while amplifying gains using tools such as leverage and stop-loss mechanisms.
- Optimize trading strategies through access to cutting-edge tools and analytics.

Pain points:

- Lack of reliable and real-time price monitoring tools to support fast decision-making.
- Difficulty in managing risk effectively due to unpredictable market volatility.
- Challenges in staying updated with the latest market news and technical indicators in real-time.
- Complexity of using multiple platforms for portfolio management and trade execution.



Overall description:

- Cryptocurrency newbie, with just six months of experience.
- He is eager to learn and focused on understanding the basics of crypto investment while cautiously experimenting with small amounts.
- His cautious nature reflects a desire to make informed decisions and build confidence in the crypto space.

Goals:

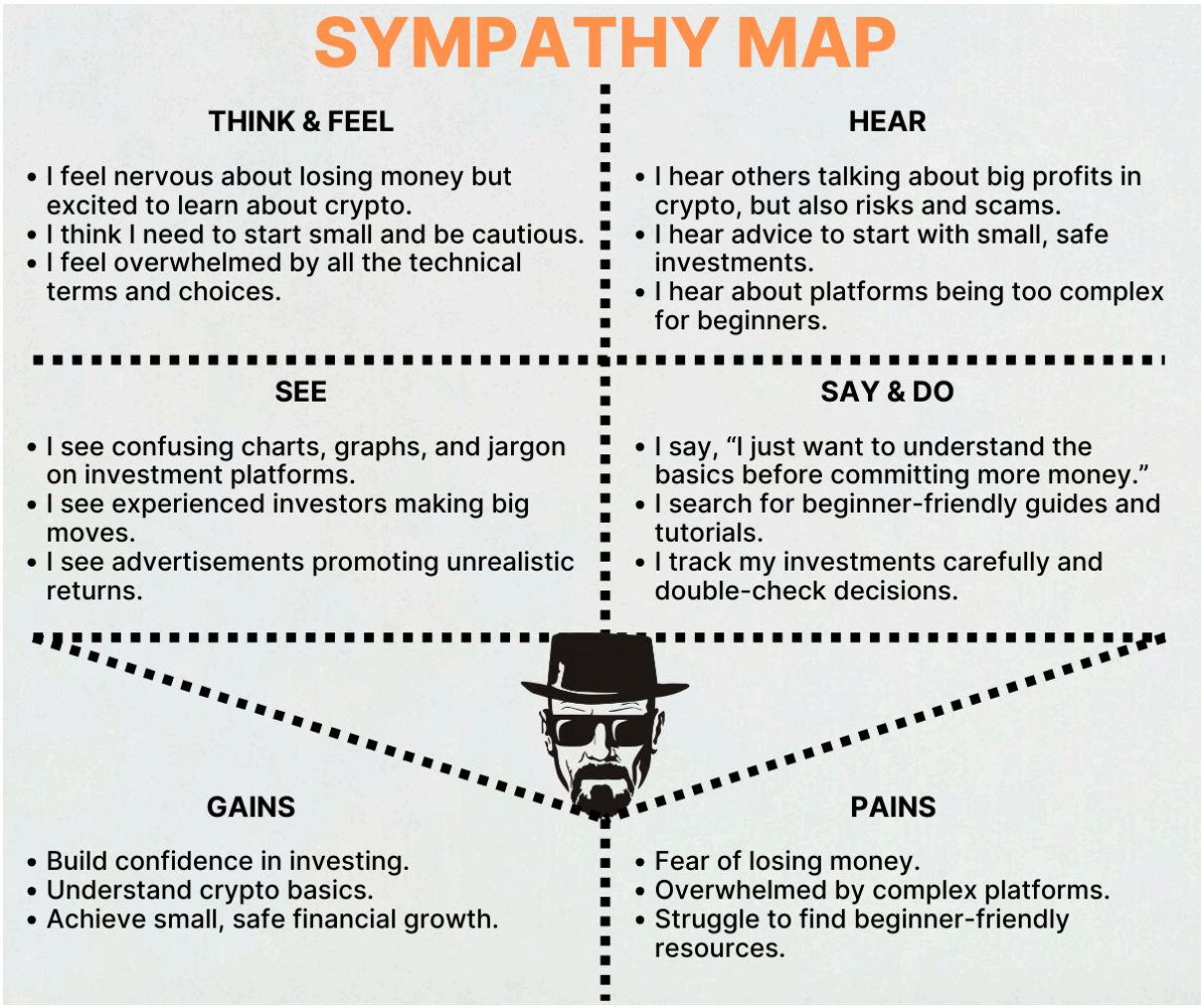
- Learn safe investing practices with minimal financial risk.
- Develop a foundational understanding of crypto concepts and trading.

- Gain the confidence needed to make more substantial investments in the future.

Pain points:

- Struggles to find beginner-friendly resources that simplify complex crypto concepts.
- Overwhelmed by technical jargon and the complexity of trading platforms.
- Difficulty in tracking and managing his small portfolio with clarity.
- Lacks a clear understanding of how to evaluate the risks and potential returns of investments.

C. Sympathy map:



D. Feature list:

a) Functional:

Wallet Integration:

Users must securely connect cryptocurrency wallets (e.g., MetaMask, Trust Wallet) to the application.

Portfolio Management:

Allows users to create, manage, and customize investment Machines with defined parameters and DCA strategies.

Automated Trading:

Executes trades automatically based on predefined DCA strategies across multiple cryptocurrencies and DEXs.

Real-Time Data Analysis:

Fetches real-time price data, displays market trends, and provides performance metrics for user portfolios.

User Interface:

Provides an intuitive dashboard for profile overviews, trade history, and clear reporting.

Notifications:

Sends in-app notifications for trade executions, market updates, and portfolio changes.

b) Non - functional:

Performance:

- Executes trades promptly within **5 seconds** of scheduled times.

Reliability:

- Ensures **99.9% uptime**, with automated backup and recovery mechanisms for data integrity.

Usability:

- Features a user-friendly interface with clear navigation and comprehensive help resources.

Compliance:

- Adheres to financial regulations and data privacy laws, ensuring transparent reporting.

III/ Produce:

A. Test cases:

1. Wallet Integration

- **Description:** Verify that the user can securely connect their cryptocurrency wallet to the application.
- **Precondition:** User has a valid cryptocurrency wallet (e.g., MetaMask, Trust Wallet) installed.
- **Test Steps:**
    1. Open the application.
    2. Navigate to the Connect Wallet screen.
    3. Select a wallet provider.
    4. Approve the connection in the wallet application.
- **Expected Result:** The wallet connects successfully, and the user is redirected to the application.

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2. Portfolio Management

- **Description:** Verify that the user can create a new investment Machine (DCA pool).
- **Precondition:** User’s wallet is connected.
- **Test Steps:**
    1. Open the Strategy screen.
    2. Select "Create New Machine."
    3. Define the investment token, target token, amount, and schedule.
    4. Confirm the creation.
    5. Approve the transaction in the wallet application.
- **Expected Result:** The new Machine is successfully created and visible in the My Machines screen.

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3. Automated Trading

- **Description:** Verify that trades are executed automatically according to the user-defined DCA strategy.
- **Precondition:** A valid Machine is created with predefined parameters.
- **Test Steps:**
    1. Wait for the scheduled trading time.
    2. Monitor the automated trade execution process.
- **Expected Result:** The trade executes as per the schedule, and the updated token balance is visible in the user's wallet.

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4. Real-Time Data Analysis

- **Description:** Verify that real-time market data and portfolio performance are displayed accurately.
- **Precondition:** User has an active Machine.
- **Test Steps:**
    1. Navigate to the Profile or My Machines screen.
    2. Review the displayed market trends, token prices, and portfolio performance metrics.
- **Expected Result:** Data is accurate and updates in real-time.

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5. User Interface

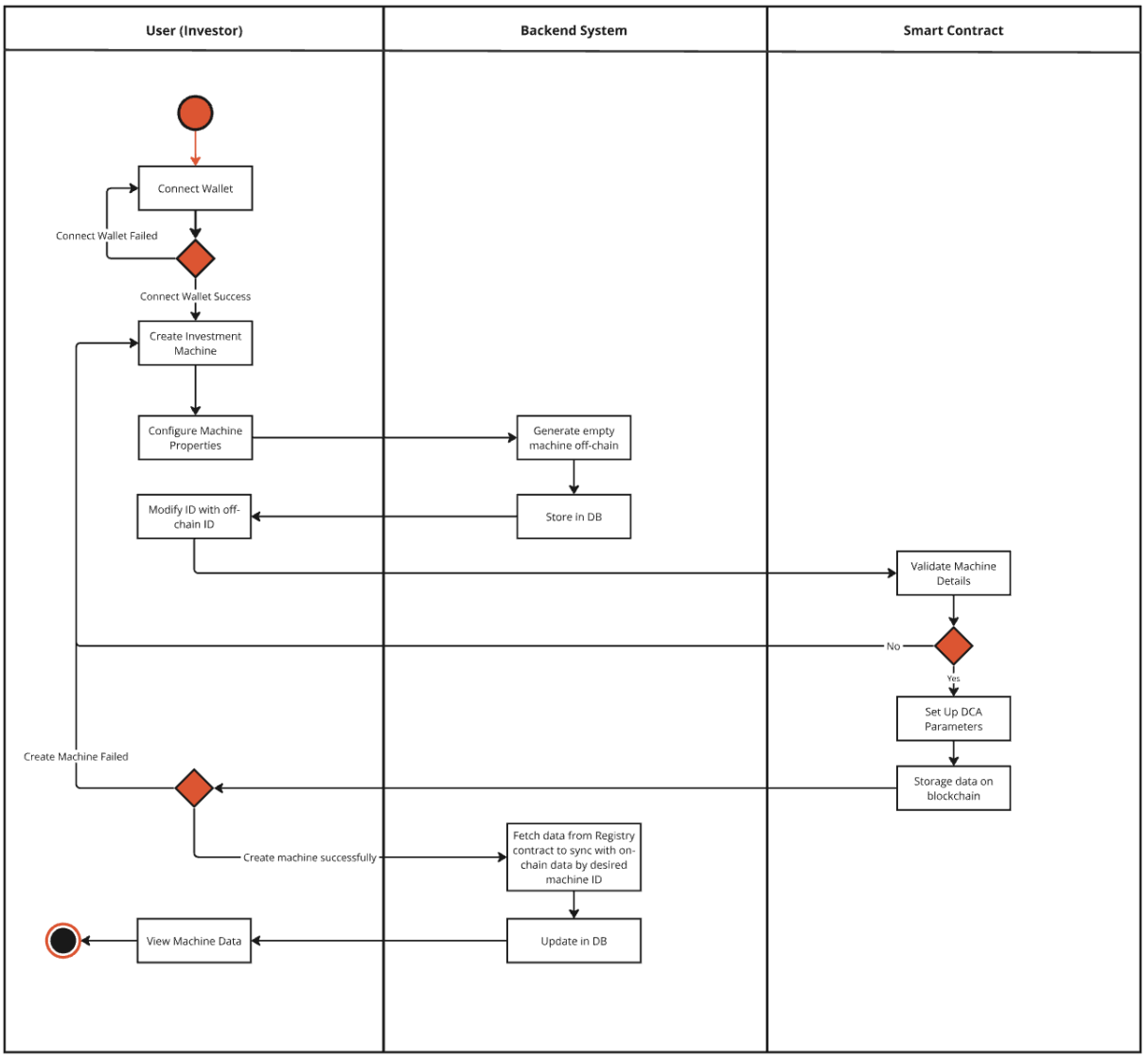
- **Test Case ID:** TC-005
- **Description:** Verify that the dashboard displays all relevant user information and navigation is intuitive.
- **Precondition:** User has created at least one Machine.
- **Test Steps:**
    1. Open the dashboard.
    2. Verify the visibility of key components: portfolio overview, trade history, and notifications.
- **Expected Result:** All information is displayed correctly, and navigation is seamless.

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6. Notifications

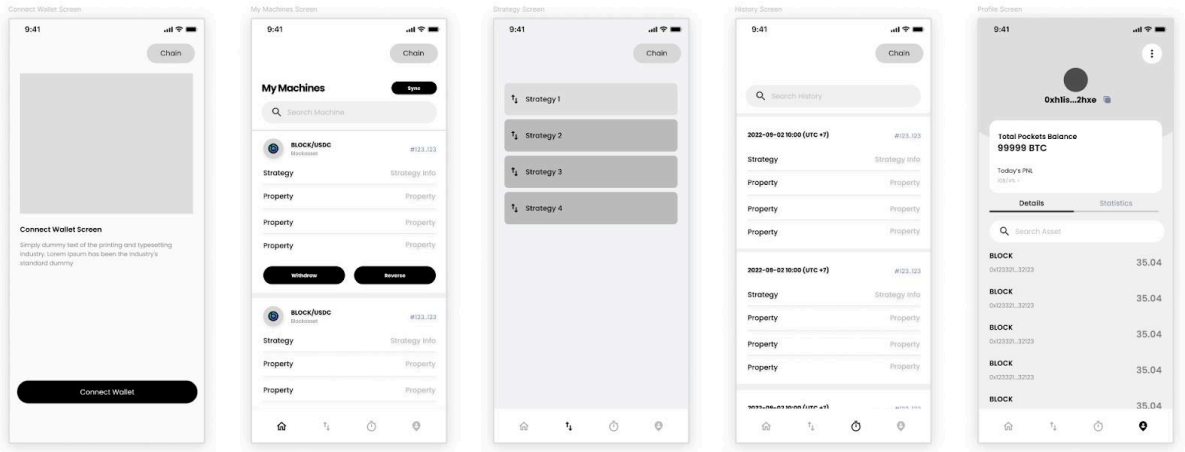
- **Description:** Verify that notifications are triggered for important events like trade executions and portfolio changes.
- **Precondition:** User has at least one active Machine.
- **Test Steps:**
    1. Wait for a scheduled trade execution or make changes to the portfolio.
    2. Check the notification center.
- **Expected Result:** Notifications are triggered and display accurate event details.

B. Flowchart:

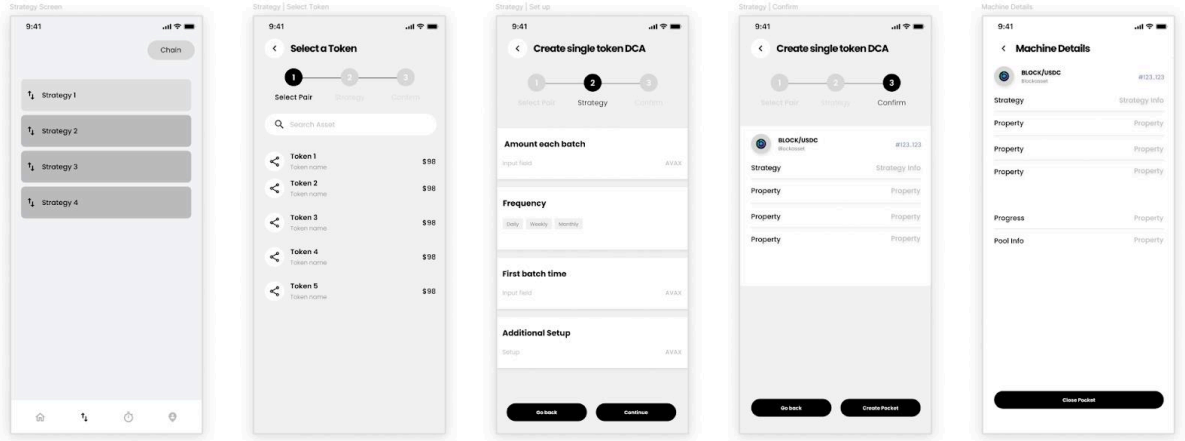


Machine creation Activity Diagram

C. Wireframe:



Main screen



Machine details - Strategy creation

PART 2 - PROTOTYPE

I/ Mobile:

[Mobile figma prototype]

II/ Web:

[Web app figma prototype]